

(Task 1 of 8) In this tutorial, we will learn about using functions as values.

0

Note: some programming languages do NOT consider functions or methods a kind of value. But many programming languages, from Python to Rust, do.

(Task 2 of 8) What is the result of running this program?

Lispy | Lispy [Run 

Scala 3

```
(deffun (inc n)      def inc(n : Int) =
  (+ n 1))          n + 1
(defvar v (mvec inc inc)) val v = (inc, inc)
((vec-ref v 0) 2)    println(v(0)(2))
```

3

2

You predicted the output correctly 🎉🎉🎉

3

`v` is bound to a vector that refers to the function `inc`. The value of `(vec-ref v 0)` is the function `inc`. So, the value of `((vec-ref v 0) 2)` is the value of `(inc 2)`, which is 3.

Click [here](#) to run this program in the Stacker.

(Task 3 of 8) What is the result of running this program?

Lispy | Lispy [Run 

Scala 3

```
(deffun (twice f x)  def twice[T](f : (T => T), x : T) =
  (f (f x)))        f(f(x))
(defun (double x)   def double(x : Int) =
  (+ x x))          x + x
(twice double 1)    println(twice(double, 1))
```

4

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You predicted the output correctly 🎉🎉🎉

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The value of `(twice double 1)` is the value of `(f (f x))`, where `f` is bound to `double` and `x` is bound to 1. So, the result is the value of `(double (double 1))`, which is 4.

Click [here](#) to run this program in the Stacker.

Lispy | 

(Task 4 of 8) What is the result of running this program?

Lispy [Run 

Pseudo

```

(deffun (inc x)    fun inc(x):
  (+ x 1))        return x + 1
(deffun (g)        end
  inc)            fun g():
(defvar f (g))     return inc
(f 10)            end
                  let f = g()
                  print(f(10))

```

error

8

The answer is 11.

9

▼ Textual explanation

You might think that (g) errors because it returns a function, and that it would not error if it returns a value of other kinds (e.g., numbers and vectors). However, it does not error. In SMoL, functions are (also) *first-class* citizens of the value world.

Click [here](#) to run this program in the Stacker.

Lispy | 

What is the result of running this program?

Lispy [Run 

Pseudo

```

(deffun (fun1)      fun fun1():
  (deffun (average x y)
    (/ (+ x y) 2))
  average)          fun average(x, y):
(defvar x (fun1))   return (x + y) / 2
(x 20 40)           end
                    return average
                    end
                    let x = fun1()
                    print(x(20, 40))

```

30

11

You predicted the output correctly 🎉🎉🎉

12

Lispy | 

(Task 5 of 8) What is the result of running this program?

Lispy [Run 

Python

```

(deffun (f)        def f():
  42)              return 42
(defvar g f)       g = f
(defvar h a)       h = a

```

<pre>def f(): return 42 g = f h = g print(h())</pre>	
42	14
You predicted the output correctly 🎉🎉🎉 This program binds <code>f</code> to a function that returns 42, and then binds <code>g</code> and <code>h</code> to that function. Finally, calling that function produces 42. Click here to run this program in the Stacker.	15
(Task 6 of 8) What did you learn about functions from these programs?	16
funcs as values	17
(Task 7 of 8) Functions are (also) <i>first-class</i> citizens of the value world. Specifically, • Variables (notably parameters) can be bound to functions, • Functions can return functions, and • Vectors can refer to functions.	18
Any feedback regarding these statements? Feel free to skip this question.	19
(You skipped the question.)	20
(Task 8 of 8) Please scroll back and select 1-3 programs that make the point above. You don't need to select <i>all</i> such programs.	21
(You selected 1 programs)	22
Okay. How does this program (13) support the point?	23
sd	24
Let's review what we have learned in this tutorial	25

Let's review what we have learned in this tutorial.

Functions are (also) *first-class* citizens of the value world. Specifically,

- Variables (notably parameters) can be bound to functions,
- Functions can return functions, and
- Vectors can refer to functions.

You have finished this tutorial 🎉🎉🎉

Please [print](#) the finished tutorial to a PDF file so you can review the content in the future. **Your instructor (if any) might require you to submit the PDF.**

Start time: 1781418859010